Qihan (Blake) Wang

(346) 391-6766 | blakew3@illinois.edu | github.com/cat-readme-txt | linkedin.com/in/qihan-blake-wang

Education

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
B.S. in Electrical Engineering — *Expected May 2030*

- **Relevant Coursework:** Differential Equations; Calculus III; Linear Algebra; Discrete Structures; Introduction to Electronics

Research Experience

EEG Motor-Imagery Classification and Signal Processing — 2026
Independent Research Project

- Built a Python/MNE EEG preprocessing pipeline implementing bandpass filtering, epoch segmentation, and frequency-domain analysis.
- Implemented time-, frequency-, and spatial-domain EEG features, including band power, Hjorth parameters, wavelet energy, CSP, and FBCSP; evaluated SVM, random forest, logistic regression, and shrinkage LDA classifiers.

Research Intern — Summer 2025
University of Houston, Department of Computer Science

- Developed Python scripts to evaluate large language model reasoning accuracy and cross-domain knowledge transfer across biology, mathematics, and law.

Projects

Chronodox Strategic Simulation Platform | *React, TypeScript, Zustand, Vite, Node.js* — 2026

- Developed a browser-based multiplayer strategy simulation with a rule-based engine, procedural map generation and spatial algorithms, and an in-browser terminal interface that exposes game actions, diagnostics, and training workflows
- Implemented REST-based client–server synchronization, built a browser-based AI self-play pipelines and trained a hybrid heuristic/candidate-based gameplay agent using 200,000+ self-play samples.

Illini Solar Car | *PCB Design, KiCad, Firmware, MCUXpresso IDE* — 2026

- Designed an LED/resistor PCB in KiCad and implemented potentiometer-controlled LED brightness in MCUXpresso IDE.

Arcade: Asteroids | *Java, Swing, Java2D, CheerpJ, Xcode* — 2025

- Developed a Java arcade game featuring AI-controlled enemies and Boids-inspired swarm behavior..
- Packaged the game as a native macOS application using Xcode and deployed a browser-playable version on GitHub Pages using CheerpJ; currently optimizing runtime performance.

Activities & Leadership

CyberPatriot | *Platinum Semifinalist, 5th Place NJROTC Semifinals* — 2025

- Hardened Linux systems in virtualized environments using Bash, system administration, and cybersecurity tools.

National Program Director, Steel City Codes — 2025–2026

- Supported computer science education through curriculum development, outreach, and program coordination; created digital outreach content generating approximately **50,000 views**.

Technical Skills

Languages: Python, C++, C, Java, Bash, TypeScript, JavaScript, SQL
Systems & Tools: Linux, Git, virtual machines, SSH, KiCad, MCUXpresso IDE
Libraries & Frameworks: Numpyt, MNE, React, Node.js, HTML, CSS